

Practical – 1 **real-world e-commerce shop UI example** that uses MediaQuery to:

- Detect **screen width** and **orientation**
- Dynamically adjust **column count**
 - 2 → phones
 - 3 → small tablets
 - 4 → large screens/web
- Change childAspectRatio based on **portrait vs landscape**

This example demonstrates how you'd build a **responsive product grid** using only MediaQuery.

```
import 'package:flutter/material.dart';
```

```
void main() {  
  runApp(MyApp());  
}
```

```
class MyApp extends StatelessWidget {  
  @override  
  Widget build(BuildContext context) {  
    return MaterialApp(  
      title: 'MediaQuery Demo',  
      debugShowCheckedModeBanner: false,  
      theme: ThemeData(primarySwatch: Colors.deepPurple),  
      home: ShopHomePage(),  
    );  
  }  
}
```

```
}
```

```
class ShopHomePage extends StatelessWidget {  
  @override  
  Widget build(BuildContext context) {  
    // Use MediaQuery to get screen information  
    double width = MediaQuery.of(context).size.width;  
    Orientation orientation = MediaQuery.of(context).orientation;  
  
    // Determine number of columns based on width  
    int crossAxisCount;  
    if (width < 600) {  
      crossAxisCount = 2; // phones  
    } else if (width < 900) {  
      crossAxisCount = 3; // small tablets  
    } else {  
      crossAxisCount = 4; // large tablets / web  
    }  
  
    return Scaffold(  
      appBar: AppBar(title: Text('Shop App')),  
      body: GridView.builder(  
        padding: const EdgeInsets.all(12),  
        gridDelegate: SliverGridDelegateWithFixedCrossAxisCount(  
          crossAxisCount: crossAxisCount,
```

```
        crossAxisSpacing: 12,  
        mainAxisSpacing: 12,  
        childAspectRatio: orientation == Orientation.portrait ? 0.75 : 1.1,  
    ),  
    itemCount: products.length,  
    itemBuilder: (context, index) => ProductCard(product: products[index]),  
),  
);  
}  
}
```

```
class Product {  
    final String title;  
    final double price;  
    final String imageUrl;  
  
    Product({required this.title, required this.price, required this.imageUrl});  
}
```

```
List<Product> products = [  
    Product(title: 'Shoes', price: 49.99, imageUrl:  
    'https://via.placeholder.com/150'),  
    Product(title: 'T-Shirt', price: 19.99, imageUrl:  
    'https://via.placeholder.com/150'),  
    Product(title: 'Watch', price: 99.99, imageUrl:  
    'https://via.placeholder.com/150'),
```

```
Product(title: 'Bag', price: 59.99, imageUrl:
'https://via.placeholder.com/150'),
Product(title: 'Headphones', price: 39.99, imageUrl:
'https://via.placeholder.com/150'),
Product(title: 'Sunglasses', price: 29.99, imageUrl:
'https://via.placeholder.com/150'),
];
```

```
class ProductCard extends StatelessWidget {
  final Product product;

  const ProductCard({required this.product});

  @override
  Widget build(BuildContext context) {
    return Card(
      shape: RoundedRectangleBorder(borderRadius:
BorderRadius.circular(10)),
      elevation: 2,
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          Expanded(
            child: ClipRRect(
              borderRadius: BorderRadius.vertical(top: Radius.circular(10)),
              child: Image.network(product.imageUrl, fit: BoxFit.cover, width:
double.infinity),
```

```
    ),  
    ),  
    Padding(  
      padding: const EdgeInsets.all(8.0),  
      child: Text(product.title, style: TextStyle(fontWeight:  
FontWeight.bold)),  
    ),  
    Padding(  
      padding: const EdgeInsets.symmetric(horizontal: 8.0),  
      child: Text('\${product.price.toStringAsFixed(2)}'),  
    ),  
  ],  
),  
);  
}  
}
```

Practical- 1 **real-world style** example of a **fully responsive Flutter UI for a News/Dashboard app** using **LayoutBuilder**.

It switches between **mobile** → **tablet** → **desktop UI** automatically based on the available width.

main.dart

```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Responsive Layout',
      home: ResponsiveDashboard(),
    );
  }
}

class ResponsiveDashboard extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
```

```

return Scaffold(
  appBar: AppBar(
    title: Text('Flutter News Dashboard'),
  ),
  body: LayoutBuilder(
    builder: (context, constraints) {
      if (constraints.maxWidth < 600) {
        // ===== MOBILE LAYOUT =====
        return _buildMobileLayout();
      } else if (constraints.maxWidth < 1024) {
        // ===== TABLET LAYOUT =====
        return _buildTabletLayout();
      } else {
        // ===== DESKTOP LAYOUT =====
        return _buildDesktopLayout();
      }
    },
  ),
);
}

// -----
// Mobile Single-column View
// -----
Widget _buildMobileLayout() {

```

```

return ListView(
  children: [
    _buildHeader(),
    _buildNewsCard(),
    _buildNewsCard(),
    _buildNewsCard(),
  ],
);
}

// -----
// Tablet Two-column View
// -----
Widget _buildTabletLayout() {
  return Row(
    children: [
      Expanded(
        flex: 2,
        child: ListView(
          children: [
            _buildHeader(),
            _buildNewsCard(),
            _buildNewsCard(),
          ],
        ),
      ),
    ],
  );
}

```

```

    ),
    Expanded(
      flex: 1,
      child: Container(
        color: Colors.blueGrey.shade50,
        child: _buildSidePanel(),
      ),
    ),
  ],
);
}

```

```

// -----
// Desktop Three-column View
// -----
Widget _buildDesktopLayout() {
  return Row(
    children: [
      Expanded(
        flex: 2,
        child: ListView(
          children: [
            _buildHeader(),
            _buildNewsCard(),
            _buildNewsCard(),

```

```

        _buildNewsCard(),
    ],
),
),
Expanded(
    flex: 1,
    child: Container(
        color: Colors.blueGrey.shade50,
        child: _buildSidePanel(),
    ),
),
Expanded(
    flex: 1,
    child: Container(
        color: Colors.blueGrey.shade100,
        child: _buildTrending(),
    ),
),
],
);
}

```

```
//-----
```

```
// Reusable Widgets
```

```
//-----
```

```
Widget _buildHeader() {  
  return Padding(  
    padding: const EdgeInsets.all(16.0),  
    child: Text(  
      'Top Headlines',  
      style: TextStyle(fontSize: 28, fontWeight: FontWeight.bold),  
    ),  
  );  
}
```

```
Widget _buildNewsCard() {  
  return Card(  
    margin: const EdgeInsets.all(16),  
    elevation: 3,  
    child: Padding(  
      padding: const EdgeInsets.all(16.0),  
      child: Column(  
        crossAxisAlignment: CrossAxisAlignment.start,  
        children: [  
          Text(  
            'News Title Here',  
            style: TextStyle(fontSize: 20, fontWeight: FontWeight.bold),  
          ),  
          SizedBox(height: 8),  
          Text(  

```

```
        'Lorem ipsum dolor sit amet consectetur adipisicing elit. '  
        'Officiis illo iusto praesentium delectus neque.',  
    ),  
  ],  
),  
),  
);  
}
```

```
Widget _buildSidePanel() {  
  return ListView(  
    padding: const EdgeInsets.all(16),  
    children: [  
      Text('Categories', style: TextStyle(fontSize: 20, fontWeight:  
FontWeight.bold)),  
      SizedBox(height: 8),  
      ListTile(title: Text('Technology')),  
      ListTile(title: Text('Science')),  
      ListTile(title: Text('Business')),  
      ListTile(title: Text('Sports')),  
    ],  
  );  
}
```

```
Widget _buildTrending() {
```

```
return ListView(  
  padding: const EdgeInsets.all(16),  
  children: [  
    Text('Trending', style: TextStyle(fontSize: 20, fontWeight:  
FontWeight.bold)),  
    SizedBox(height: 8),  
    ListTile(title: Text('Flutter 3.22 Released!')),  
    ListTile(title: Text('AI takes over mobile apps')),  
    ListTile(title: Text('SpaceX launches new rocket')),  
  ],  
);  
}
```

Step-3 Run your App

Flutter Run

Output:

